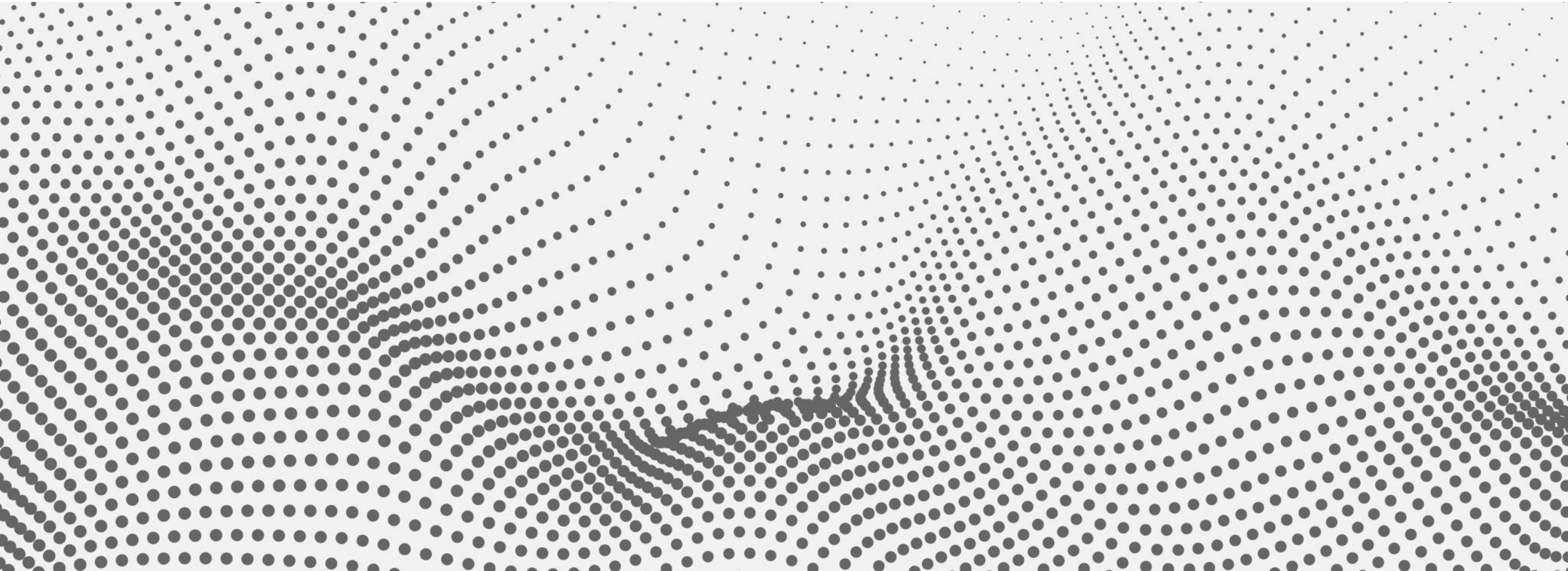
NORMALIZATION



RETAIL SLIP

- Design a schema for the retail slip using the normalization technique up to BCNF and create a Relational Data Model.
- Submit your Relational Data Model as a PDF.

https://ehopper.com/articles/5-important-points-to-consider-when-choosing-a-pos-printer/

Some address
мпав, fd, 23rf
Phone: +333333333333
SALE

Order #:	010900269558
Sold To:	Walk-In
Order Date:	03/31/17
Order Time:	1:03:23 PM
Sales Person:	Admin Admin
Register:	1
Order type:	Quick Sale

Red Banana is in business	@22.23 x 3	$66.96
Apple	@5.99 x 5.55	$33.24
Discount Sale		($5.95)

Subtotal	$100.23
Discount	($5.95)
12 %	$11.32
Total	**$105.60**

Cash Tendered	$105.60
CHANGE DUE	**$0.00**

Special package

30% off your next order –
if you order through us with code LOCAL!

UNF

ORDERS
order_no PK
sold_type
order_date
order_time
staff_id
staff_name
reg_no
order_type
disc_amt
tax_amt
total_amt
item_no*
prod_no
prod_name
unit_price
qty
item_amt

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item_no*
prod_no
prod_name
unit_price
qty
item_amt

1NF

ORDER_HEAD
order_no PK
sold_type
order_date
order_time
staff_id
staff_name
reg_no
order_type
disc_amt
tax_amt
total_amt

ORDER_DETAILS
order_no PFK
item_no PK
prod_no
prod_name
unit_price
qty
item_amt

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tax_amt
total_amt

ORDER_DETAILS
order_no PFK
item_no PK
prod_no
prod_name
unit_price
qty
item_amt

2NF

ORDER_HEAD
order_no PK
sold_type
order_date
order_time
staff_id
staff_name
reg_no
order_type
disc_amt
tax_amt
total_amt

ORDER_DETAILS
order_no PFK
item_no PK
prod_no
prod_name
unit_price
qty
item_amt

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order_no PK
sold_type
order_date
order_time
staff_id
staff_name
reg_no
order_type
disc_amt
tax_amt
total_amt

ORDER_DETAILS
order_no PFK
item_no PK
prod_no
prod_name
unit_price
qty
item_amt

3NF

ORDER_HEAD
order_no PK
order_date
order_time
staff_id FK
reg_no
sold_type
order_type
disc_amt
tax_amt
total_amt

ORDER_DETAILS
order_no PFK
item_no PK
prod_no
unit_price
qty
item_amt

STAFF
staff_id PK
staff_name

PRODUCTS
prod_no PK
prod_name

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order_no PK
sold_type
order_date
order_time
staff_id
staff_name
reg_no
order_type
disc_amt
tax_amt
total_amt

ORDER_DETAILS
order_no PFK
item_no PK
prod_no
prod_name
unit_price
qty
item_amt

2NF

ORDER_HEAD
order_no PK
sold_type
order_date
order_time
staff_id
staff_name
reg_no
order_type
disc_amt
tax_amt
total_amt

ORDER_DETAILS
order_no PFK
item_no PK
prod_no
prod_name
unit_price
qty
item_amt

3NF

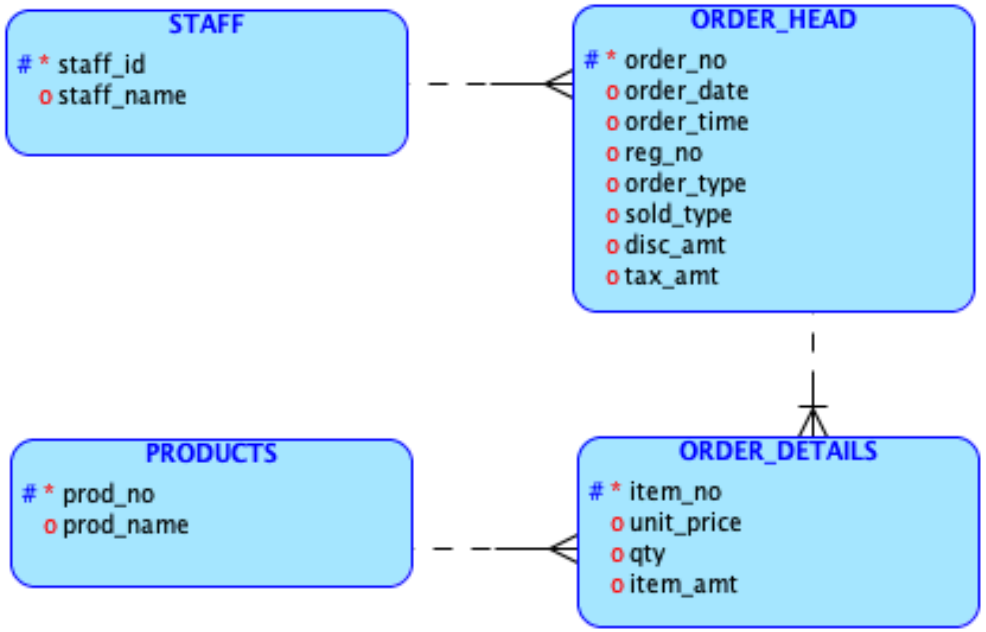
ORDER_HEAD
order_no PK
order_date
order_time
staff_id FK
reg_no
sold_type
order_type
disc_amt
tax_amt
total_amt

ORDER_DETAILS
order_no PFK
item_no PK
prod_no
unit_price
qty
item_amt

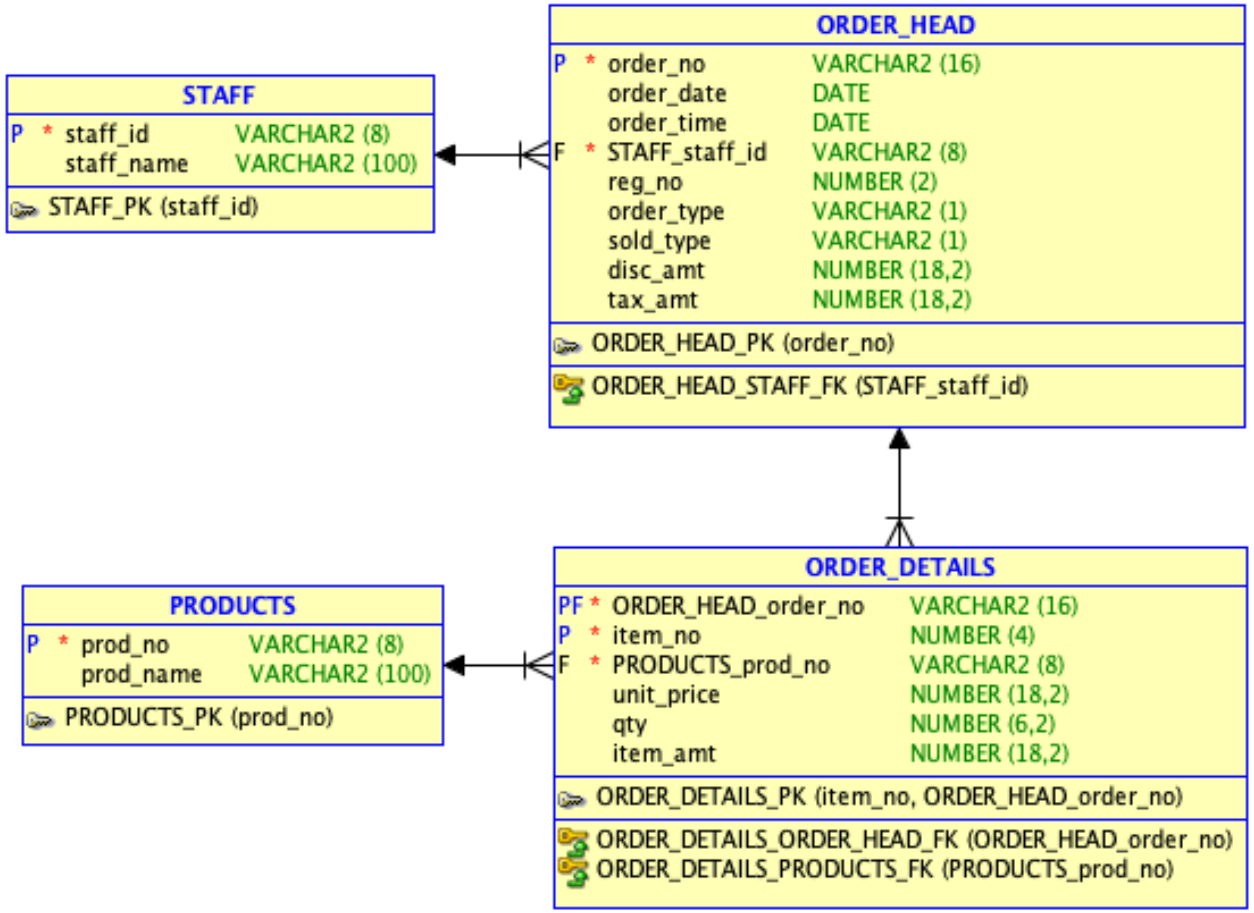
STAFF
staff_id PK
staff_name

PRODUCTS
prod_no PK
prod_name

Logical Model



Relational Model (physical)



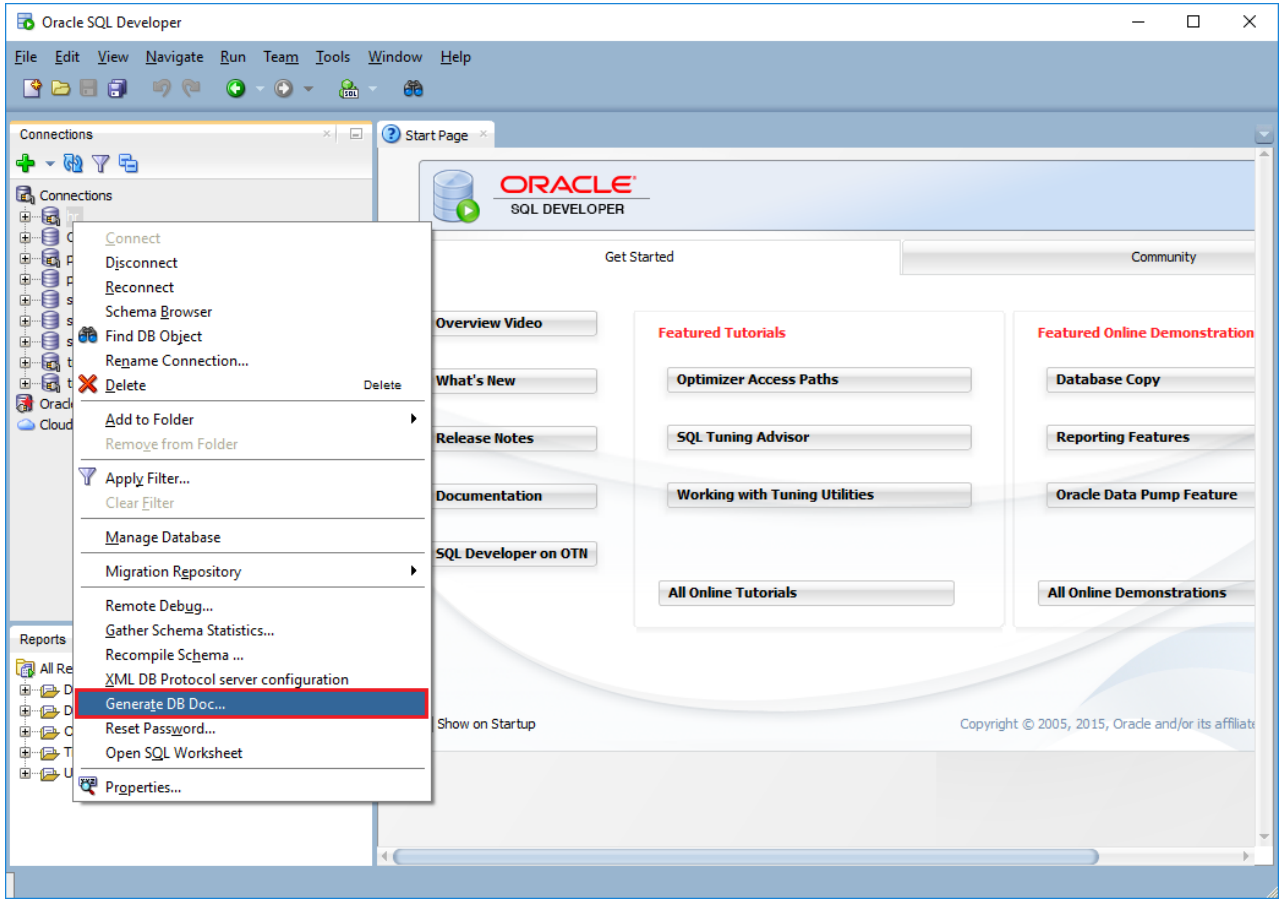
DDL File Editor - Oracle Database 11g

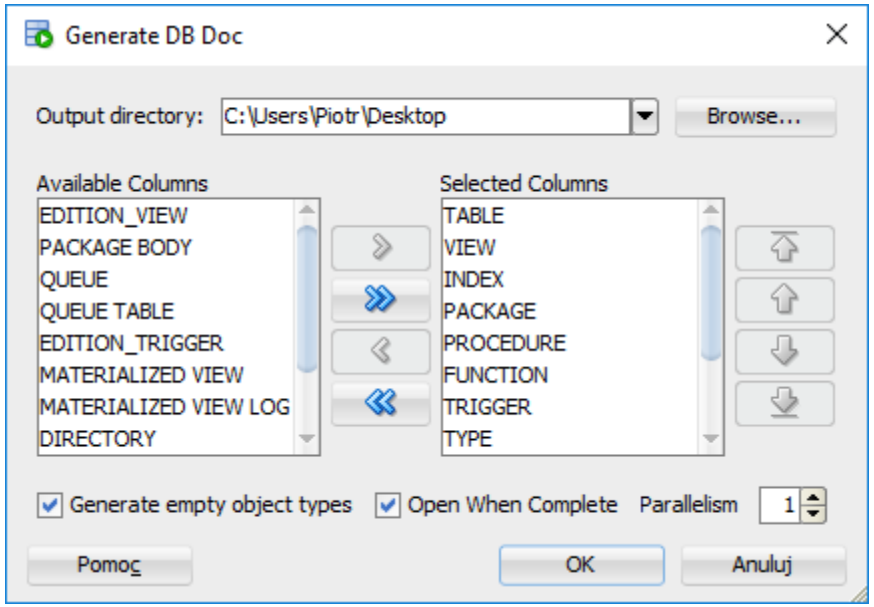
Oracle Database 11g | Relational_1 | Generate | Clear

```
-- Generated by Oracle SQL Developer Data Modeler 20.2.0.167.1538
--   at:        2023-09-05 14:56:27 ICT
--   site:      Oracle Database 11g
--   type:      Oracle Database 11g
-- predefined type, no DDL - MDSYS.SDO_GEOMETRY
-- predefined type, no DDL - XMLTYPE

CREATE TABLE order_details (
    order_head_order_no  VARCHAR2(16) NOT NULL,
    item_no              NUMBER(4) NOT NULL,
    products_prod_no     VARCHAR2(8) NOT NULL,
    unit_price           NUMBER(18, 2),
    qty                  NUMBER(6, 2),
    item_amt             NUMBER(18, 2)
);

ALTER TABLE order_details ADD CONSTRAINT order_details_pk PRIMARY KEY ( item_no,
                                                                        order_head_order_no );

CREATE TABLE order_head (
    order_no        VARCHAR2(16) NOT NULL,
    order_date      DATE,
    order_time      DATE,
    staff_staff_id  VARCHAR2(8) NOT NULL,
    reg_no          NUMBER(2),
    order_type      VARCHAR2(1),
    sold_type       VARCHAR2(1),
    disc_amt        NUMBER(18, 2),
    tax_amt         NUMBER(18, 2)
);

COMMENT ON COLUMN order_head.order_type IS
    'Q: Quick Sale';

COMMENT ON COLUMN order_head.sold_type IS
    'M: Member
N: Non-member';

ALTER TABLE order_head ADD CONSTRAINT order_head_pk PRIMARY KEY ( order_no );
```

Save | Find | Close | Help

GENERATE A DATA DICTIONARY





https://dataedo.com/kb/tools/oracle-sql-developer/how-to-export-data-dictionary

GENERATE A DATA DICTIONARY

HR

- Tables
- Views
- Indexes
- Packages
- Procedures
- Functions
- Triggers
- Types
- Sequences
- Synonyms
- Database Links
- Public Database Links

Tables

- COUNTRIES
- DEPARTMENTS
- EMPLOYEES
- JOBS
- JOB_HISTORY
- LOCATIONS
- REGIONS

EMPLOYEES

Columns | Constraints | Grants | Statistics | Triggers | Dependencies | Details | Partitions | Indexes

COLUMN_NAME	DATA_TYPE	NULLABLE	DATA_DEFAULT	COLUMN_ID	COMMENTS
EMPLOYEE_ID	NUMBER(6,0)	No	null	1	Primary key of employees table.
FIRST_NAME	VARCHAR2(20 BYTE)	Yes	null	2	First name of the employee. A not null column.
LAST_NAME	VARCHAR2(25 BYTE)	No	null	3	Last name of the employee. A not null column.
EMAIL	VARCHAR2(25 BYTE)	No	null	4	Email id of the employee
PHONE_NUMBER	VARCHAR2(20 BYTE)	Yes	null	5	Phone number of the employee; includes country code and area code
HIRE_DATE	DATE	No	null	6	Date when the employee started on this job. A not null column.
JOB_ID	VARCHAR2(10 BYTE)	No	null	7	Current job of the employee; foreign key to job_id column of the jobs table. A not null column.
SALARY	NUMBER(8,2)	Yes	50000	8	Monthly salary of the employee. Must be greater than zero (enforced by constraint emp_salary_min)
COMMISSION_PCT	NUMBER(2,2)	Yes	null	9	Commission percentage of the employee; Only employees in sales department elgible for commission percentage
MANAGER_ID	NUMBER(6,0)	Yes	null	10	Manager id of the employee; has same domain as manager_id in departments table. Foreign key to employee_id column of employees table. (useful for reflexive joins and CONNECT BY query)
DEPARTMENT_ID	NUMBER(4,0)	Yes	null	11	Department id where employee works; foreign key to department_id column of the departments table

https://dataedo.com/kb/tools/oracle-sql-developer/how-to-export-data-dictionary